Where does revenue leak between confirmation and cash?

Working-capital diagnostic · Order-to-Cash · SQL + Power BI

Orders	Order value (EUR)	Actual DSO	Contract terms
91,746	**€652.9m**	**44.1 days**	**54.5 days**

The question

Average DSO looks healthy: paid invoices clear in 44 days against 54 days of contractual terms. The leak is not the average. It is the late, blocked, and disputed cases. This brief names the top three leakage points, a process owner for each, and the cash-acceleration case.

Top three leakage points (by cash locked)

1	**Collection lag after payment terms** Owner: AR Collections Lead	**€96.9m** 56,046 orders · €388.8m involved
2	**Dispute / rework loops** Owner: AR Dispute Manager (Sales contributing)	**€4.5m** 22,474 orders · €178.4m involved
3	**Holds (delivery / credit-style)** Owner: Credit Manager / Order Management	**€1.9m** 6,482 orders · mostly operational, not credit limit

Cash-acceleration case

If extra days versus payment terms were cut to zero, expected cash released is about €97 million. At a 6% cost of capital that is about €5.8 million of annual interest. This is a scenario, not a forecast. A 10-day cut on the same late/open book is a more realistic programme target.

Formula: cash locked = order value (EUR) × extra days / 365. Extra days are defined per leak (late vs terms; dispute vs clean peers; hold vs no-hold peers; goods-issue to invoice).

What to do

1 Collections

Work overdue open AR and chronic late payers first in EMEA and Power Tools. Do not manage to the healthy company-average DSO.

2 Disputes

Stop invoice errors at order entry (price, material). Give Sales a named SLA with AR on open dispute loops.

3 Holds

Do not treat every hold as a credit failure. True credit-limit blocks are almost none (4 orders). Clear operational blocks faster; keep credit as a control.

Billing (watch)

18,915 orders and €128m flagged, but only €1.1m cash locked — extra days are short. Owner: Billing / Order-to-Invoice Lead.

Where it sits

Region: EMEA holds most value and most collection cash. AMER next. APAC is smaller in value but slower on invoice-to-cash.
Segment: External customers (~€94m collection cash) versus internal (~€3m).
Business area: Power Tools, then Agriculture, then Building Materials.

O2C path in SQL (vw_o2c_case_timeline)

One row per sales order. Columns read left to right as the process: Create Sales Order → Confirmed Delivery Date → Issue Goods → Create Invoice → Invoice Due Date → Clear Invoice. Holding and rework dates sit beside the happy path so missing next documents stay visible. Query: sql/04_o2c_timeline.sql.

```
SELECT sales_order_id, ts_create_sales_order, ts_confirmed_delivery,
       ts_issue_goods, ts_create_invoice, ts_invoice_due, ts_clear_invoice
FROM vw_o2c_case_timeline;
```

Power BI report — four pages

File: powerbi/O2C_Leakage.pbix. Export a visual pack from Power BI Desktop: File → Export → Export to PDF (or Print → Microsoft Print to PDF). Combine with this brief for applications.

1 Executive

KPIs (orders, €653m, DSO 44 vs terms 54). Bar of cash locked by leak. Owner table. Cash by region. Slicers do not move the company snapshot; they move the region chart.

2 Process

Events by stage group (fulfilment largest, hold smallest). Average days: goods→invoice ~0, invoice→cash ~46, confirm→cash ~41. Happy-path share vs extra steps.

3 Leakage

Order counts on flags. Cash by leak. Delivery-block reasons (No block dominates; credit limit is rare). Collection cash by business area (Power Tools).

4 Segment & region

Matrix region × customer group. Collection cash by area and region. Customer ranking by collection cash locked.

Method

Celonis Academy trained the process vocabulary (case, activity, timestamp, variant, waiting time). Analysis data is not a Celonis extract.
Sources: QPR Order-to-Cash event log and case attributes; ECB euro reference rates. Joins: Sales Order ID (91,746/91,746); date + sales-org currency to ECB.
Python loads files into SQLite. Star schema, leak flags, and DSO marts are SQL (sql/02_build_star.sql, sql/03_metrics.sql).

Limits

Public sample process data, not a live SAP client extract. Amounts treated as sales-org currency then converted to EUR. Classic AR-stock DSO is not used. Leak cash figures can overlap — do not add them as unique euros.

Rebuild locally

1. Clone the repo. Download OTC_Events_900k.csv, OTC_Cases_92k.csv, and ECB eurofxref-hist.csv into data/raw/ (URLs in README).

2. python sql/01_load_and_build.py

3. In DB Browser, run sql/03_metrics.sql and sql/04_o2c_timeline.sql if you rebuilt only part of the pipeline.

4. Open powerbi/O2C_Leakage.pbix in Power BI Desktop.

5. Optional: File → Export → PDF for the four report pages; keep this PDF as the narrative.

Credit

Author of analysis, SQL model, Power BI structure, and this brief: Manikanta Engalligi.
Event log samples: QPR ProcessAnalyzer. FX: European Central Bank. Process-intelligence training: Celonis Academy.